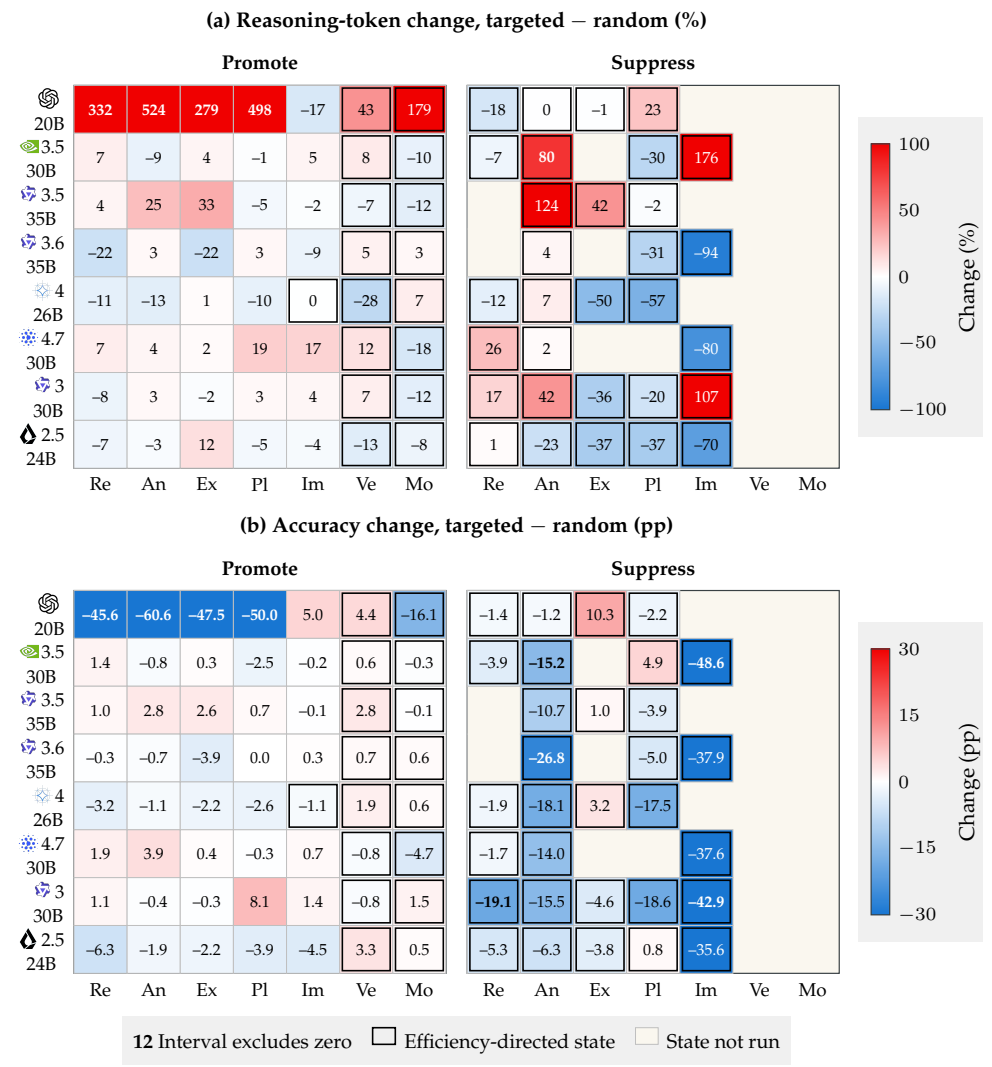

(a) Reasoning-token change, targeted − random (%)
Promote
Suppress
20B
3.5 30B
3.5 35B
3.6 35B
4 26B
4.7 30B
3 30B
2.5 24B
332 524 279 498 −17 43 179
−18 0 −1 23
7 −9 4 −1 5 8 −10
−7 80 −30 176
4 25 33 −5 −2 −7 −12
124 42 −2
−22 3 −22 3 −9 5 3
4 −31 −94
−11 −13 1 −10 0 −28 7
−12 7 −50 −57
7 4 2 19 17 12 −18
26 2 −80
−8 3 −2 3 4 7 −12
17 42 −36 −20 107
−7 −3 12 −5 −4 −13 −8
1 −23 −37 −37 −70
Re An Ex Pl Im Ve Mo
Re An Ex Pl Im Ve Mo
100
50
0
−50
−100
Change (%)
(b) Accuracy change, targeted − random (pp)
Promote
Suppress
20B
3.5 30B
3.5 35B
3.6 35B
4 26B
4.7 30B
3 30B
2.5 24B
−45.6 −60.6 −47.5 −50.0 5.0 4.4 −16.1
−1.4 −1.2 10.3 −2.2
1.4 −0.8 0.3 −2.5 −0.2 0.6 −0.3
−3.9 −15.2 4.9 −48.6
1.0 2.8 2.6 0.7 −0.1 2.8 −0.1
−10.7 1.0 −3.9
−0.3 −0.7 −3.9 0.0 0.3 0.7 0.6
−26.8 −5.0 −37.9
−3.2 −1.1 −2.2 −2.6 −1.1 1.9 0.6
−1.9 −18.1 3.2 −17.5
1.9 3.9 0.4 −0.3 0.7 −0.8 −4.7
−1.7 −14.0 −37.6
1.1 −0.4 −0.3 8.1 1.4 −0.8 1.5
−19.1 −15.5 −4.6 −18.6 −42.9
−6.3 −1.9 −2.2 −3.9 −4.5 3.3 0.5
−5.3 −6.3 −3.8 0.8 −35.6
Re An Ex Pl Im Ve Mo
Re An Ex Pl Im Ve Mo
30
15
0
−15
−30
Change (pp)
12 Interval excludes zero
Efficiency-directed state
State not run